Artificial Intelligence Model Practice Paper

Artificial Intelligence - Demo paper - 2026 edition - 50 marks

Self-created practice paper for NoteNest, not an official university paper.

Section A - Short answers (5 x 4 marks)

Q1. Define a rational agent and list the types of agent programs. **[4 marks]**

Q2. What makes a heuristic admissible? **[4 marks]**

Q3. Differentiate between BFS and DFS in terms of completeness and memory. **[4 marks]**

Q4. Differentiate between supervised and unsupervised learning with an example each. **[4 marks]**

Q5. Define overfitting and list two ways to reduce it. **[4 marks]**

Section B - Long answers (2 x 15 marks)

Q6. Explain the A* algorithm with the evaluation function f(n) = g(n) + h(n). Prove or argue that A* is optimal with an admissible heuristic. **[15 marks]**

Q7. A classifier produces TP=45, FP=5, FN=15, TN=35. Compute accuracy, precision, recall and F1. Comment on which metric matters for a spam filter. **[15 marks]**

Answer hints

- Q7: accuracy 0.80, precision 0.90, recall 0.75, F1 about 0.818.
- Q7: for a spam filter, high precision avoids blocking real mail.